

# Umesh A

Bengaluru, India | umesharumugam6@gmail.com | 7892629935 | [GitHub](#) | [LinkedIn](#)

| Course Name | Institute Name                          | University Name | Year of Passing | Score Details (%/CGPA) | Course Type |
|-------------|-----------------------------------------|-----------------|-----------------|------------------------|-------------|
| 10th        | Bangalore Education Society             | KSEEB           | 2021            | 93.9%                  | Full-time   |
| 12th        | Mysore Education Society Kishore kendra | KSEAB           | 2023            | 93.3%                  | Full-time   |
| UG          | Dayananda Sagar College of Engineering  | VTU             | 2027            | 9.01                   | Full-time   |

## EXPERIENCE

### Full Stack Development & AI Intern, Imaginext Innovations

Aug 2026 - Present

- Develop full-stack features using React and Node.js while integrating AI/ML components into existing product workflows.
- Design and implement AI-driven functionalities within a broader application architecture, ensuring robust software testing and deployment.

### Full Stack Development & AI Intern, TrackLeaf

May 2026 - Aug 2026

- Engineered full-stack features and integrated AI/ML components to automate complex data processing tasks, improving system efficiency.
- Collaborated with engineering teams in an agile environment to design and implement robust, scalable software solutions.

### Data Analyst Intern, Axlero Innovative Solutions

May 2026 - Jul 2026

- Clean and analyze large datasets using SQL and Excel to surface trends and build interactive dashboards for stakeholder communication.
- Translate analytical insights into actionable business recommendations through cross-functional team collaboration.

### Cybersecurity Intern, Guardex Infosec and Technology

Mar 2026 - Jun 2026

- Perform penetration testing on web applications to identify vulnerabilities (injection, broken authentication) and document findings in structured remediation reports.
- Collaborate with the security team on threat analysis and validate security fixes through rigorous re-testing processes.

## TECHNICAL SKILLS

- AI/ML Frameworks:** TensorFlow, Keras, OpenCV, YOLOv8, Scikit-learn
- Core AI/ML Concepts:** Machine Learning, Computer Vision, NLP, Deep Learning, CNN, Model Deployment
- Data Tools:** Pandas, NumPy, Jupyter
- Programming Languages:** Python, Java, JavaScript
- Web & Backend:** Flask, Node.js, Express.js, REST APIs
- Security Foundations:** Network Security, Vulnerability Assessment, Penetration Testing, IDS/IPS, Threat Analysis
- Tools & Platforms:** Git, GitHub, VS Code, Streamlit

## PROJECTS

### VeriShield AI - Fraud Detection Platform

Python, Machine Learning, Blockchain 

- Built an AI fraud detection platform using behavioral ML models that trigger real-time hashing of high-risk account details onto a blockchain ledger.
- Implemented pre-transaction blockchain lookups to automatically block matches and prevent repeat fraud in real time.

### Brain Tumor Detection using CNN

Python, TensorFlow, Keras, CNN 

- Designed a CNN-based deep learning model to classify brain MRI scans into tumor and non-tumor categories
- Co-authored and published an IEEE research paper on this medical image analysis project

### Traffic Light Detection using YOLOv8 & Gemini Assistant

Python, YOLOv8, Computer Vision, Gemini API 

- Built a real-time computer vision system using YOLOv8 to detect traffic lights and classify signal colors
- Integrated a Gemini-based voice assistant to announce signal status, improving accessibility for visually impaired users

### AI-Powered Phishing Detection Browser Extension

Python, Machine Learning, Browser Extension APIs 

- Developed a real-time phishing detection browser extension using machine learning and URL-based feature extraction to flag malicious websites
- Designed an interactive warning system that alerts users instantly upon detecting suspicious or phishing URLs

## ACHIEVEMENTS

- Secured 4th Rank at National Level in the Math Talent Exam (2022)
- Secured 6th Rank at State Level in the Math Talent Exam (2020)
- Published an IEEE research paper on CNN-based brain tumor detection